

1. Setup e Imports

```
# Instalar SAELens si no está disponible:
# !pip install sae-lens

import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint_dir
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4, suppress=True)

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Dispositivo: {device}')
```

Dispositivo: cuda

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints
NAMING_META = {
    'variante': 'topk',          # Arquitectura del SAE
    'tecnica': 'batch-topk',    # BatchTopK
    'capa': 6,                  # Layer del modelo OthelloGPT
    'juegos': 1500,             # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',
    'experiment_base_path': os.path.abspath('../../experiments'),
    'extras': {
        'expansion': 8,        # d_sae / d_in = 8192 / 512
        'k': 50,               # número de features activas L0 promedio    k
        'epochs': 1000,
        'patience': 20
    }
}

print('\nMetadata de naming:')
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

Metadata de naming:

```
variante: topk
tecnica: batch-topk
capa: 6
juegos: 1500
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
```

```
extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
```

```

) if NAMING_META.get('extras') else None

# get_plots_dir crea el directorio automáticamente
plots_dir = Path(get_plots_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
))

print(f'extras_id: {extras_id}')
print(f'plots_dir: {plots_dir}')

```

```

extras_id: ex1
plots_dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_topk\1500games\ex1\plots

```

3. Configuración del SAE con SAELens

Configuración usando `LanguageModelSAERunnerConfig` de SAELens con `architecture='batchtopk'`.

Arquitectura BatchTopK: - **BatchTopK activation:** calcula un threshold global por batch para que el **promedio** de features activas por muestra sea k (L0 promedio k , pero varía por muestra) - **Sin coeficiente de sparsidad:** la sparsity se controla directamente con k - **Loss = MSE únicamente** (sin penalización L1)

Parámetro	L1 (standard)	TopK	BatchTopK
Sparsity	<code>sparsity_coef</code> (soft)	k (hard, por muestra)	k (promedio del batch)
L0	Variable	Fijo = k por muestra	Variable por muestra, promedio k
Loss	$\text{MSE} + \cdot \text{L1}$	MSE	MSE

Nota sobre el runner: `LanguageModelSAETrainingRunner(cfg).run()` requiere que Othello-GPT esté cargado vía `TransformerLens`. Las celdas siguientes entrenan el SAE directamente sobre los archivos `.npy` pre-extraídos usando la arquitectura BatchTopK definida en el config.

```

NUM_EPOCHS = NAMING_META['extras']['epochs']

cfg = {
    'architecture':      'batchtopk',
    'd_in':              512,
    'd_sae':             4096,
    'k':                 50,
    'lr':                1e-3,
    'train_batch_size_tokens': 4096,
    'hook_name':         'blocks.5.hook_resid_post',
}

cfg_summary = pd.DataFrame({
    'Parámetro SAELens': ['architecture', 'd_in', 'd_sae', 'k (L0 promedio)', 'hook_name', 'lr'],
    'Valor': [cfg['architecture'], cfg['d_in'], cfg['d_sae'], cfg['k'],
              cfg['hook_name'], cfg['lr']]
})

print('Configuración SAELens:')
print(cfg_summary.to_string(index=False))

```

Configuración SAELens:

Parámetro SAELens	Valor
architecture	batchtopk
d_in	512
d_sae	4096
k (L0 promedio)	50
hook_name	blocks.5.hook_resid_post
lr	0.001

4. Dataset de Activaciones (Train/Validation Split 80-20%)

Activaciones pre-extraídas con `sae/activations/run_extraction.py` desde la capa `blocks.5.hook_resid_post`.

```

class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""
    def __init__(self, activations_path):
        self.activations = np.load(activations_path)
    def __len__(self):
        return len(self.activations)
    def __getitem__(self, idx):

```

```

        return torch.tensor(self.activations[idx], dtype=torch.float32)

activations_file = Path(f"../../activations/data/layer{NAMING_META['capa']}-{NAMING_META['id']}")
full_dataset = ActivationsDataset(activations_file)

# Split 80-20 reproducible
train_size = int(0.8 * len(full_dataset))
val_size    = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset, [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

train_loader = DataLoader(train_dataset, batch_size=cfg["train_batch_size_tokens"])
val_loader   = DataLoader(val_dataset,   batch_size=cfg["train_batch_size_tokens"])

dataset_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validación', 'Total'],
    'Muestras': [f'{len(train_dataset):,}', f'{len(val_dataset):,}', f'{len(full_dataset):,}'],
    'Shape':     [str(full_dataset.activations.shape), '', ''],
})
print(dataset_info.to_string(index=False))

```

```

Conjunto Muestras      Shape
Train    70,800 (88500, 512)
Validación 17,700
Total     88,500

```

5. Modelo: BatchTopK SAE

Implementación de la arquitectura BatchTopK de SAELens.

Diferencia clave vs ReLU + L1 y vs TopK estándar: - El encoder proyecta a `d_sae=8192` dimensiones (pre-activación) - La activación **BatchTopK** calcula un threshold global del batch (k-ésimo valor promediado); cada muestra puede tener más o menos de `k` features activas - L0 es **variable por muestra**, pero el **promedio del batch** `k` (soft constraint a nivel de batch) - No hay `sparsity_coef` ni cosine annealing - El decoder se normaliza a norma unitaria después de cada paso del optimizador

```

class BatchTopKSAE(nn.Module):
    """

```

```

Sparse Autoencoder con activación BatchTopK.
Equivalente a SAELens architecture='batchtopk'.

Encoder      : d_in -> d_sae (lineal + pre-bias centering)
Activación   : BatchTopK (mantiene el promedio de k features activas por batch)
Decoder      : d_sae -> d_in (columnas con norma unitaria)
"""
def __init__(self, d_in: int, d_sae: int, k: int, k_aux: int = 512):
    super().__init__()
    self.d_in = d_in
    self.d_sae = d_sae
    self.k = k
    self.k_aux = k_aux

    self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
    self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
    self.b_dec = nn.Parameter(torch.zeros(d_in))

    nn.init.kaiming_uniform_(self.W_enc)
    nn.init.kaiming_uniform_(self.W_dec)
    self.W_dec.data = self.W_dec.data / self.W_dec.data.norm(dim=-1, keepdim=True)
    self.register_buffer('num_batches_not_active', torch.zeros(d_sae))
    self.n_batches_to_dead = 10 # latente muerto si inactivo por 10 batches seguidos

def encode(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """Proyecta a espacio latente con activación BatchTopK"""
    x_cent = x - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    return self._topk_activation(pre_acts, threshold=threshold)

def _topk_activation(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """BatchTopK: selecciona exactamente k * batch_size activaciones usando scatter."""
    if threshold is None:
        acts_topk = torch.topk(x.flatten(), self.k * x.shape[0], dim=-1)
        return (
            torch.zeros_like(x.flatten())
            .scatter(-1, acts_topk.indices, acts_topk.values)
            .reshape(x.shape)
        )
    return torch.where(x > threshold, x, torch.zeros_like(x))

def decode(self, hidden: torch.Tensor) -> torch.Tensor:

```

```

        return hidden @ self.W_dec + self.b_dec

def preprocess_input(self, x: torch.Tensor):
    x_mean = x.mean(dim=-1, keepdim=True)
    x = x - x_mean
    x_std = x.std(dim=-1, keepdim=True)
    x = x / (x_std + 1e-5)
    return x, x_mean, x_std

def postprocess_output(self, x_reconstruct: torch.Tensor,
                        x_mean: torch.Tensor, x_std: torch.Tensor) -> torch.Tensor:
    return x_reconstruct * x_std + x_mean

def forward(self, x: torch.Tensor, threshold=None):
    x_norm, x_mean, x_std = self.preprocess_input(x)
    x_cent = x_norm - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    hidden = self._topk_activation(pre_acts, threshold=threshold)
    recon = self.postprocess_output(self.decode(hidden), x_mean, x_std)
    return recon, hidden, pre_acts

def auxiliary_loss(self, x: torch.Tensor, x_reconstruct: torch.Tensor, acts: torch.Tensor):
    """L_aux del repo oficial: reconstruye el residual con top-k_aux dead latents."""
    dead_features = self.num_batches_not_active >= self.n_batches_to_dead
    if dead_features.sum() == 0:
        return torch.tensor(0.0, device=x.device)
    residual = x.float() - x_reconstruct.float()
    acts_topk_aux = torch.topk(
        acts[:, dead_features],
        min(self.k_aux, int(dead_features.sum().item()))),
        dim=-1,
    )
    acts_aux = torch.zeros_like(acts[:, dead_features]).scatter(
        -1, acts_topk_aux.indices, acts_topk_aux.values
    )
    x_reconstruct_aux = acts_aux @ self.W_dec[dead_features]
    return (1/32) * (x_reconstruct_aux.float() - residual.float()).pow(2).mean()

@torch.no_grad()
def update_inactive_features(self, acts):
    self.num_batches_not_active += (acts.sum(0) == 0).float()
    self.num_batches_not_active[acts.sum(0) > 0] = 0

```

```

@torch.no_grad()
def estimate_threshold(self, loader, device, n_batches: int = 10) -> float:
    """Estima un threshold estable promediando sobre múltiples batches
    (independiente del batch actual - para usar en validación/inferencia)."""
    self.eval()
    thresholds = []
    for i, batch in enumerate(loader):
        if i >= n_batches:
            break
        x = batch.to(device)
        x_norm, _, _ = self.preprocess_input(x)
        x_cent = x_norm - self.b_dec
        pre_acts = F.relu(x_cent @ self.W_enc)
        mins = []
        for sample_idx in range(pre_acts.shape[0]):
            pos = pre_acts[sample_idx][pre_acts[sample_idx] > 0]
            if pos.numel() > 0:
                mins.append(pos.min().item())
        t = float(np.mean(mins)) if mins else 0.0
        thresholds.append(t)
    return float(np.mean(thresholds))

@torch.no_grad()
def make_decoder_weights_and_grad_unit_norm(self):
    """Proyecta el gradiente del decoder y renormaliza - igual que el repo oficial."""
    W_dec_normed = self.W_dec / self.W_dec.norm(dim=-1, keepdim=True)
    W_dec_grad_proj = (self.W_dec.grad * W_dec_normed).sum(-1, keepdim=True) * W_dec_normed
    self.W_dec.grad -= W_dec_grad_proj
    self.W_dec.data = W_dec_normed

# Crear modelo desde parámetros del cfg SAELens
model = BatchTopKSAE(d_in=cfg['d_in'], d_sae=cfg['d_sae'], k=cfg['k']).to(device)
num_params = sum(p.numel() for p in model.parameters())

model_info = pd.DataFrame({
    'Componente': [
        'Input', 'Encoder (W_enc)', 'Activación', 'Hidden activos', 'Decoder (W_dec)', 'Output'
    ],
    'k_aux': 'Total params'
},
    'Dimensión': [
        f'{cfg["d_in"]}',

```



```

        f'{cfg["d_in"]} -> {cfg["d_sae"]}',
        f'BatchTopK (k={cfg["k"]})',
        f'{cfg["d_sae"]} (solo {cfg["k"]} activos en promedio)',
        f'{cfg["d_sae"]} -> {cfg["d_in"]}',
        f'{cfg["d_in"]}',
        f'{model.k_aux}',
        f'{num_params:,'}
    ]
})
print(model_info.to_string(index=False))

```

Componente	Dimensión
Input	512
Encoder (W_enc)	512 -> 4096
Activación	BatchTopK (k=50)
Hidden activos	4096 (solo 50 activos en promedio)
Decoder (W_dec)	4096 -> 512
Output	512
k_aux	512
Total params	4,194,816

6. Función de Pérdida BatchTopK

Loss = MSE(reconstrucción)

- **MSE**: error de reconstrucción (igual que en L1)
- **Sin L1 penalty**: la sparsity ya está controlada por el threshold global de BatchTopK
- No hay `sparsity_coef` ni annealing — el único hiperparámetro de sparsity es `k` (promedio del batch)

```

ALPHA_AUX = 1 / 32 # peso de la pérdida auxiliar (arXiv:2412.06410)

def loss_function(x, reconstruction, acts, model):
    """
    Pérdida BatchTopK: MSE + alpha * L_aux (arXiv:2412.06410).
    - MSE      : error de reconstrucción
    - L_aux    : reconstruye el residual con las top-k_aux features para
                  evitar features muertas (alpha = 1/32)

    Args:
        x: Activaciones originales
    """

```

```

        reconstruction: Activaciones reconstruidas
        acts:           Pre-activaciones del encoder
        model:          Instancia de BatchTopKSAE
Returns:
    total_loss (scalar)
"""
mse    = F.mse_loss(reconstruction, x)
l_aux  = model.auxiliary_loss(x, reconstruction, acts)
return mse + l_aux

optimizer = torch.optim.Adam(model.parameters(), lr=cfg["lr"], betas=(0.9, 0.99))

optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning rate', 'betas', 'k (L0 promedio k)', 'k_aux', 'alpha_aux'],
    'Valor':      ['Adam', cfg["lr"], '(0.9, 0.99)', cfg["k"], model.k_aux, f'{ALPHA_AUX:.5f}']
})
print(optim_info.to_string(index=False))

```

	Parámetro	Valor
	Optimizador	Adam
	Learning rate	0.001
	betas	(0.9, 0.99)
k (L0 promedio k)		50
	k_aux	512
	alpha_aux	0.03125 (1/32)

7. Loop de Entrenamiento

```

PATIENCE = NAMING_META['extras']['patience']

save_dir = Path('./saved_model')
save_dir.mkdir(exist_ok=True)
best_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='best',
    extras_id=extras_id
)

```

```

)
BEST_CKPT = save_dir / best_model_name

history = {
    'train_mse_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_mse_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
}

best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'Iniciando entrenamiento: {NUM_EPOCHS} épocas máximo')
print(f'Early stopping: patience={PATIENCE}')
print(f'k = {cfg["k"]} features activas (L0 promedio k por batch)')
print(f'k_aux = {model.k_aux} | alpha_aux = {ALPHA_AUX:.5f} (1/32)')
print(f'Best checkpoint: {BEST_CKPT}')
print('=' * 70)

for epoch in range(NUM_EPOCHS):

    # TRAIN
    model.train()
    t_mse = t_l0 = t_frac = 0.0

    for batch in train_loader:
        x = batch.to(device)
        optimizer.zero_grad()
        recon, hidden, pre_acts = model(x)
        total_loss = loss_function(x, recon, pre_acts, model)
        total_loss.backward()
        model.make_decoder_weights_and_grad_unit_norm()
        optimizer.step()
        model.update_inactive_features(hidden)

    with torch.no_grad():
        t_mse += F.mse_loss(recon, x).item()
        t_l0 += (hidden > 0).float().sum(dim=1).mean().item()

```

```

        t_frac += (hidden > 0).float().mean().item()

n = len(train_loader)
history['train_mse_loss'].append(t_mse / n)
history['train_l0_sparsity'].append(t_l0 / n)
history['train_fraction_active'].append(t_frac / n)

# VALIDATION (sin threshold - igual que training, L0 k)
model.eval()
v_mse = v_l0 = v_frac = 0.0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)
        recon, hidden, _ = model(x)
        v_mse += F.mse_loss(recon, x).item()
        v_l0 += (hidden > 0).float().sum(dim=1).mean().item()
        v_frac += (hidden > 0).float().mean().item()

nv = len(val_loader)
history['val_mse_loss'].append(v_mse / nv)
history['val_l0_sparsity'].append(v_l0 / nv)
history['val_fraction_active'].append(v_frac / nv)

print(f"Época {epoch+1:3d}/{NUM_EPOCHS} | "
      f"Train MSE: {history['train_mse_loss'][-1]:.6f} | "
      f"Val MSE: {history['val_mse_loss'][-1]:.6f} | "
      f"Train L0: {history['train_l0_sparsity'][-1]:.1f} | "
      f"Val L0: {history['val_l0_sparsity'][-1]:.1f}")

# Early stopping
if history['val_mse_loss'][-1] < best_val_mse:
    best_val_mse = history['val_mse_loss'][-1]
    best_epoch = epoch + 1
    epochs_no_improve = 0

torch.save(model.state_dict(), BEST_CKPT)
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],

```

```

        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=cfg,
        history=history,
        optimizer=optimizer
    )
    print(f' Mejor modelo guardado (época {best_epoch} | Val MSE: {best_val_mse:.6f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= PATIENCE:
        print(f'\nEarly stopping en época {epoch+1} (mejor: época {best_epoch})')
        break

# Restaurar mejor modelo
model.load_state_dict(torch.load(BEST_CKPT, weights_only=True))
print(f'\nMejor época: {best_epoch} | Mejor Val MSE: {best_val_mse:.6f}')

# Estimar threshold de inferencia al final del entrenamiento
inference_threshold = model.estimate_threshold(train_loader, device, n_batches=10)
print(f'Threshold de inferencia estimado: {inference_threshold:.4f}')

```

Iniciando entrenamiento: 1000 épocas máximo

Early stopping: patience=20

k = 50 features activas (L0 promedio k por batch)

k_aux = 512 | alpha_aux = 0.03125 (1/32)

Best checkpoint: saved_model\sae_topk_batch-topk_l6_1500g_ex1_best.pt

=====

Época 1/1000 | Train MSE: 0.724390 | Val MSE: 0.547802 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt

Mejor modelo guardado (época 1 | Val MSE: 0.547802)

Época 2/1000 | Train MSE: 0.476600 | Val MSE: 0.416132 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt

Mejor modelo guardado (época 2 | Val MSE: 0.416132)

Época 3/1000 | Train MSE: 0.371053 | Val MSE: 0.340689 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt

Mejor modelo guardado (época 3 | Val MSE: 0.340689)

Época 4/1000 | Train MSE: 0.312924 | Val MSE: 0.301889 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 4 | Val MSE: 0.301889)
Época 5/1000 | Train MSE: 0.280788 | Val MSE: 0.279625 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 5 | Val MSE: 0.279625)
Época 6/1000 | Train MSE: 0.260666 | Val MSE: 0.265369 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 6 | Val MSE: 0.265369)
Época 7/1000 | Train MSE: 0.246828 | Val MSE: 0.254766 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 7 | Val MSE: 0.254766)
Época 8/1000 | Train MSE: 0.236188 | Val MSE: 0.247120 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 8 | Val MSE: 0.247120)
Época 9/1000 | Train MSE: 0.227784 | Val MSE: 0.240871 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 9 | Val MSE: 0.240871)
Época 10/1000 | Train MSE: 0.220591 | Val MSE: 0.235707 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 10 | Val MSE: 0.235707)
Época 11/1000 | Train MSE: 0.214311 | Val MSE: 0.231442 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 11 | Val MSE: 0.231442)
Época 12/1000 | Train MSE: 0.208885 | Val MSE: 0.227732 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 12 | Val MSE: 0.227732)
Época 13/1000 | Train MSE: 0.204028 | Val MSE: 0.224767 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 13 | Val MSE: 0.224767)
Época 14/1000 | Train MSE: 0.199656 | Val MSE: 0.221977 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 14 | Val MSE: 0.221977)

Época 15/1000 | Train MSE: 0.195595 | Val MSE: 0.219414 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 15 | Val MSE: 0.219414)
 Época 16/1000 | Train MSE: 0.191743 | Val MSE: 0.217395 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 16 | Val MSE: 0.217395)
 Época 17/1000 | Train MSE: 0.188111 | Val MSE: 0.215472 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 17 | Val MSE: 0.215472)
 Época 18/1000 | Train MSE: 0.185254 | Val MSE: 0.213715 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 18 | Val MSE: 0.213715)
 Época 19/1000 | Train MSE: 0.181975 | Val MSE: 0.211947 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 19 | Val MSE: 0.211947)
 Época 20/1000 | Train MSE: 0.178889 | Val MSE: 0.210411 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 20 | Val MSE: 0.210411)
 Época 21/1000 | Train MSE: 0.176092 | Val MSE: 0.209145 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 21 | Val MSE: 0.209145)
 Época 22/1000 | Train MSE: 0.173724 | Val MSE: 0.208089 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 22 | Val MSE: 0.208089)
 Época 23/1000 | Train MSE: 0.171665 | Val MSE: 0.207067 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 23 | Val MSE: 0.207067)
 Época 24/1000 | Train MSE: 0.169497 | Val MSE: 0.206270 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 24 | Val MSE: 0.206270)
 Época 25/1000 | Train MSE: 0.167534 | Val MSE: 0.205249 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt

Mejor modelo guardado (época 25 | Val MSE: 0.205249)
 Época 26/1000 | Train MSE: 0.165523 | Val MSE: 0.204751 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 26 | Val MSE: 0.204751)
 Época 27/1000 | Train MSE: 0.163942 | Val MSE: 0.203751 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 27 | Val MSE: 0.203751)
 Época 28/1000 | Train MSE: 0.162247 | Val MSE: 0.203113 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 28 | Val MSE: 0.203113)
 Época 29/1000 | Train MSE: 0.160862 | Val MSE: 0.202680 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 29 | Val MSE: 0.202680)
 Época 30/1000 | Train MSE: 0.159521 | Val MSE: 0.201885 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 30 | Val MSE: 0.201885)
 Época 31/1000 | Train MSE: 0.158156 | Val MSE: 0.201545 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 31 | Val MSE: 0.201545)
 Época 32/1000 | Train MSE: 0.156886 | Val MSE: 0.200973 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 32 | Val MSE: 0.200973)
 Época 33/1000 | Train MSE: 0.155663 | Val MSE: 0.200570 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 33 | Val MSE: 0.200570)
 Época 34/1000 | Train MSE: 0.154768 | Val MSE: 0.200270 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 34 | Val MSE: 0.200270)
 Época 35/1000 | Train MSE: 0.153921 | Val MSE: 0.199818 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 35 | Val MSE: 0.199818)
 Época 36/1000 | Train MSE: 0.152772 | Val MSE: 0.199488 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt

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 Mejor modelo guardado (época 36 | Val MSE: 0.199488)
 Época 37/1000 | Train MSE: 0.151863 | Val MSE: 0.199200 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 37 | Val MSE: 0.199200)
 Época 38/1000 | Train MSE: 0.151113 | Val MSE: 0.199043 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 38 | Val MSE: 0.199043)
 Época 39/1000 | Train MSE: 0.150346 | Val MSE: 0.198742 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 39 | Val MSE: 0.198742)
 Época 40/1000 | Train MSE: 0.149525 | Val MSE: 0.198506 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 40 | Val MSE: 0.198506)
 Época 41/1000 | Train MSE: 0.148740 | Val MSE: 0.198330 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 41 | Val MSE: 0.198330)
 Época 42/1000 | Train MSE: 0.148051 | Val MSE: 0.198222 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 42 | Val MSE: 0.198222)
 Época 43/1000 | Train MSE: 0.147451 | Val MSE: 0.198074 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 43 | Val MSE: 0.198074)
 Época 44/1000 | Train MSE: 0.146919 | Val MSE: 0.197980 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 44 | Val MSE: 0.197980)
 Época 45/1000 | Train MSE: 0.146372 | Val MSE: 0.197944 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 45 | Val MSE: 0.197944)
 Época 46/1000 | Train MSE: 0.145933 | Val MSE: 0.197791 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_l6_1500g_ex1_best.pt
 Mejor modelo guardado (época 46 | Val MSE: 0.197791)
 Época 47/1000 | Train MSE: 0.145333 | Val MSE: 0.197640 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 47 | Val MSE: 0.197640)
Época 48/1000 | Train MSE: 0.144655 | Val MSE: 0.197513 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 48 | Val MSE: 0.197513)
Época 49/1000 | Train MSE: 0.144156 | Val MSE: 0.197434 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 49 | Val MSE: 0.197434)
Época 50/1000 | Train MSE: 0.143775 | Val MSE: 0.197436 | Train L0: 50.0 | Val L0: 50.0
Época 51/1000 | Train MSE: 0.143434 | Val MSE: 0.197406 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 51 | Val MSE: 0.197406)
Época 52/1000 | Train MSE: 0.142999 | Val MSE: 0.197316 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 52 | Val MSE: 0.197316)
Época 53/1000 | Train MSE: 0.142504 | Val MSE: 0.197180 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 53 | Val MSE: 0.197180)
Época 54/1000 | Train MSE: 0.142083 | Val MSE: 0.197106 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 54 | Val MSE: 0.197106)
Época 55/1000 | Train MSE: 0.141715 | Val MSE: 0.197051 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 55 | Val MSE: 0.197051)
Época 56/1000 | Train MSE: 0.141391 | Val MSE: 0.197144 | Train L0: 50.0 | Val L0: 50.0
Época 57/1000 | Train MSE: 0.141037 | Val MSE: 0.196980 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 57 | Val MSE: 0.196980)
Época 58/1000 | Train MSE: 0.140701 | Val MSE: 0.197069 | Train L0: 50.0 | Val L0: 50.0
Época 59/1000 | Train MSE: 0.140381 | Val MSE: 0.196997 | Train L0: 50.0 | Val L0: 50.0
Época 60/1000 | Train MSE: 0.140021 | Val MSE: 0.196947 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\1500games\sae_topk_batch-topk_16_1500g_ex1_best.pt
Mejor modelo guardado (época 60 | Val MSE: 0.196947)

Época	61/1000		Train MSE: 0.139776		Val MSE: 0.197015		Train L0: 50.0		Val L0: 50.0
Época	62/1000		Train MSE: 0.139469		Val MSE: 0.197134		Train L0: 50.0		Val L0: 50.0
Época	63/1000		Train MSE: 0.139219		Val MSE: 0.197194		Train L0: 50.0		Val L0: 50.0
Época	64/1000		Train MSE: 0.138968		Val MSE: 0.197084		Train L0: 50.0		Val L0: 50.0
Época	65/1000		Train MSE: 0.138750		Val MSE: 0.197288		Train L0: 50.0		Val L0: 50.0
Época	66/1000		Train MSE: 0.138510		Val MSE: 0.197082		Train L0: 50.0		Val L0: 50.0
Época	67/1000		Train MSE: 0.138206		Val MSE: 0.197208		Train L0: 50.0		Val L0: 50.0
Época	68/1000		Train MSE: 0.137980		Val MSE: 0.197052		Train L0: 50.0		Val L0: 50.0
Época	69/1000		Train MSE: 0.137664		Val MSE: 0.197107		Train L0: 50.0		Val L0: 50.0
Época	70/1000		Train MSE: 0.137452		Val MSE: 0.197074		Train L0: 50.0		Val L0: 50.0
Época	71/1000		Train MSE: 0.137229		Val MSE: 0.197152		Train L0: 50.0		Val L0: 50.0
Época	72/1000		Train MSE: 0.137013		Val MSE: 0.197105		Train L0: 50.0		Val L0: 50.0
Época	73/1000		Train MSE: 0.136778		Val MSE: 0.197238		Train L0: 50.0		Val L0: 50.0
Época	74/1000		Train MSE: 0.136584		Val MSE: 0.197177		Train L0: 50.0		Val L0: 50.0
Época	75/1000		Train MSE: 0.136390		Val MSE: 0.197366		Train L0: 50.0		Val L0: 50.0
Época	76/1000		Train MSE: 0.136312		Val MSE: 0.197519		Train L0: 50.0		Val L0: 50.0
Época	77/1000		Train MSE: 0.136084		Val MSE: 0.197475		Train L0: 50.0		Val L0: 50.0
Época	78/1000		Train MSE: 0.135867		Val MSE: 0.197448		Train L0: 50.0		Val L0: 50.0
Época	79/1000		Train MSE: 0.135678		Val MSE: 0.197397		Train L0: 50.0		Val L0: 50.0
Época	80/1000		Train MSE: 0.135493		Val MSE: 0.197621		Train L0: 50.0		Val L0: 50.0

Early stopping en época 80 (mejor: época 60)

Mejor época: 60 | Mejor Val MSE: 0.196947

Threshold de inferencia estimado: 0.0015

8. Guardar Modelo

```
# Guardar checkpoint final
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': {
        'architecture': cfg["architecture"],
        'd_in': cfg["d_in"],
        'd_sae': cfg["d_sae"],
        'k': cfg["k"],
        'lr': cfg["lr"],
        'hook_name': cfg["hook_name"],
    },
    'history': history,
```

```

}

final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras_id=extras_id
)
checkpoint_path = save_dir / final_model_name
torch.save(checkpoint, checkpoint_path)
print(f'Modelo guardado: {checkpoint_path}')

save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=cfg,
    history=history,
    optimizer=optimizer
)

```

Modelo guardado: saved_model\sae_topk_batch-topk_l6_1500g_ex1_final.pt
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_batch-topk_l6_1500g_ex1_final.pt

WindowsPath('C:/Users/Esposa/Documents/Repos/sae-othello-gpt/layer_06/models/sae_topk/sae_topk_batch-topk_l6_1500g_ex1_final.pt')

9. Métricas básicas del modelo

```

import json

metrics_dir = get_metrics_dir(

```

```

NAMING_META['experiment_base_path'],
NAMING_META['variante'],
NAMING_META['tecnica'],
NAMING_META['capa'],
NAMING_META['juegos'],
extras_id=extras_id
)

training_metrics = {
    # Calidad de reconstrucción
    'val_mse_final': history['val_mse_loss'][-1],
    'val_mse_best': best_val_mse,
    'val_l0_final': history['val_l0_sparsity'][-1],
    'val_fraction_active_final': history['val_fraction_active'][-1],
    # Contexto del entrenamiento
    'best_epoch': best_epoch,
    'total_epochs': len(history['train_mse_loss']),
    'train_mse_final': history['train_mse_loss'][-1],
    'train_l0_final': history['train_l0_sparsity'][-1],
    # Hiperparámetros TopK
    'architecture': cfg["architecture"],
    'k': cfg["k"],
    'd_sae': cfg["d_sae"],
    'd_in': cfg["d_in"],
    'expansion_factor': cfg["d_sae"] // cfg["d_in"],
    'hook_name': cfg["hook_name"],
}

os.makedirs(metrics_dir, exist_ok=True)
metrics_path = Path(metrics_dir) / 'training_metrics.json'
with open(metrics_path, 'w') as f:
    json.dump(training_metrics, f, indent=2)

print(f'Métricas guardadas: {metrics_path}')
print(pd.DataFrame(
    list(training_metrics.items()), columns=['Métrica', 'Valor']
).to_string(index=False))

```

```

Métricas guardadas: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae\
topk_topk1500games\ex1\metrics\training_metrics.json

```

Métrica	Valor
val_mse_final	0.197621

val_mse_best	0.196947
val_l0_final	50.0
val_fraction_active_final	0.012207
best_epoch	60
total_epochs	80
train_mse_final	0.135493
train_l0_final	50.0
architecture	batchtopk
k	50
d_sae	4096
d_in	512
expansion_factor	8
hook_name	blocks.5.hook_resid_post

10. Visualización de Resultados

```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# MSE Loss
axes[0].plot(history['train_mse_loss'], label='Train', color='steelblue', alpha=0.8)
axes[0].plot(history['val_mse_loss'], label='Validación', color='tomato', alpha=0.8)
if best_epoch > 0:
    axes[0].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7,
                    label=f'Best (época {best_epoch})')
axes[0].set_title('MSE Loss (Reconstrucción)')
axes[0].set_xlabel('Época')
axes[0].set_ylabel('MSE')
axes[0].legend()
axes[0].grid(True, alpha=0.3)

# L0 sparsity (para BatchTopK el promedio del batch k, pero varía por muestra)
axes[1].plot(history['train_l0_sparsity'], label='Train', color='steelblue', alpha=0.8)
axes[1].plot(history['val_l0_sparsity'], label='Validación', color='tomato', alpha=0.8)
axes[1].axhline(cfg["k"], color='orange', linestyle='--', alpha=0.8, label=f'k={cfg["k"]} (p
axes[1].set_title(f'L0 Sparsity (promedio del batch k={cfg["k"]})')
axes[1].set_xlabel('Época')
axes[1].set_ylabel('Features activas (promedio)')
axes[1].legend()
axes[1].grid(True, alpha=0.3)

# Fracción activa
```

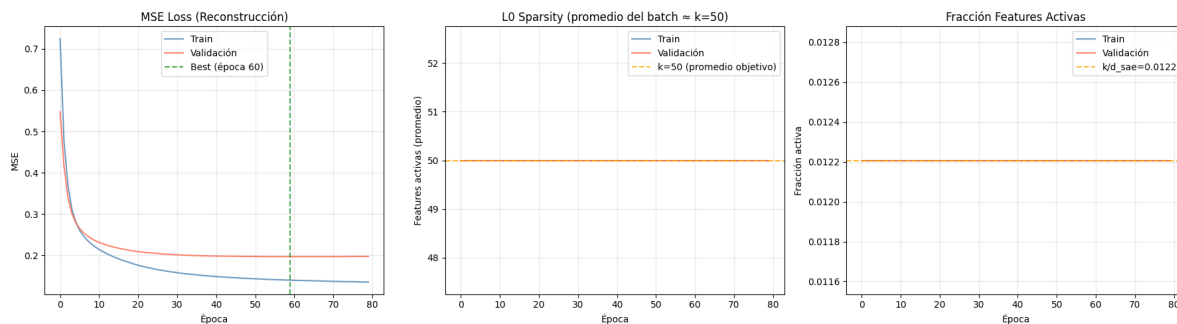
```

axes[2].plot(history['train_fraction_active'], label='Train', color='steelblue', alpha=0.8)
axes[2].plot(history['val_fraction_active'], label='Validación', color='tomato', alpha=0.8)
expected_frac = cfg["k"] / cfg["d_sae"]
axes[2].axhline(expected_frac, color='orange', linestyle='--', alpha=0.8,
                label=f'k/d_sae={expected_frac:.4f}')
axes[2].set_title('Fracción Features Activas')
axes[2].set_xlabel('Época')
axes[2].set_ylabel('Fracción activa')
axes[2].legend()
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(plots_dir / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('Gráficas guardadas exitosamente')
print('='*50)

```



=====
Gráficas guardadas exitosamente
=====

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```

import subprocess

exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_batch-topk.ipynb'
output_path = exp_dir / pdf_name

print(f'Generando PDF con Quarto...')
print(f'Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f'PDF generado exitosamente: {output_path}')
else:
    print(f'Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk_topk\1500games\ex1\sae_topk_batch-topk_l6_1500g_ex1_training.pdf

PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\lay
topk_topk\1500games\ex1\sae_topk_batch-topk_16_1500g_ex1_training.pdf